



Potential of meeting electricity needs of off-grid community with mini-grid solar systems

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ABSTRACT

The abundance of renewable energy in Ghana can play an important role in the rural electrification program rolled out by the Government of Ghana to promote energy that everyone can assess for improved living standards. It has the potential to meet the objectives of the energy sector which include; 10% renewable energy in the total generation mix, minimize the adverse effects of energy production on the environment, reduce poverty, and improve the socio-economic development of the country, mainly, in rural communities, creating community-based employment, etc. In this study, Nkrankrom was selected as a case study to evaluate the possibility of meeting their energy needs with a solar mini-grid. To determine the energy demand of the community, a structured questionnaire was used to collect data. The HOMER software was used to perform the financial viability of utilizing the solar resource available. The result shows that it is possible to meet the energy demand of the community from the solar resource available. The proposed system configuration included PV/Battery/Converter with a Levelized Cost of Energy of \$0.107/kWh compared to \$0.124/kWh (using \$1 = GHC4.98 rate), if the area is to be connected to the national grid. The breakeven distance or Electric Distance Limit (EDL) between standalone mini-grid and grid extension in this analysis was found to be 1.11 km. The study also revealed that solar mini-grid could have an immense benefit to the community both economically and socially such as improve the standard of living as well as meeting the rural development objectives of Ghana.

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Introduction

Sustainable development has been outlined in the long-term policies and strategies in several African countries, where growing energy consumption stands as a critical factor [1]. Securing energy supply remains a major global energy challenge

Abbreviation: s: CFL, Compact fluorescent lamp; PV, Photovoltaics; NPC, Net Present Cost; LCOE, levelised Cost of Energy; W, Wattage; NES, National Electrification Scheme; kW, Kilowatt; kWh, Kilowatt hour; NASA, National Aeronautics and Space Administration; SHS, Solar Home Systems; HOMER, Hybrid Optimization for Electric Renewables.

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towards meeting growing demand, providing universal access to energy services, diversification of energy generation resources, and contributing to climate change [2,3]. Economic growth and energy use, as well as the level of development, are seen to have a high degree of relationship [3,4]. Greater energy access leads to improvement in standards of living [5]. This suggests that people living in off-grid communities do not experience economic growth. Some of these communities are located very far from the national grid, while the sort of settlements is very scattered, making it economically unviable to transmit electricity from the grid to these areas. These communities are often characterized by well-recognized challenges [6]. These challenges include; distance from national or regional electricity grids, difficult terrain such as rivers or jungles, harsh weather conditions, etc. [7–13]. The challenge of grid electricity extension to geographically remote and sparsely populated off-grid communities has always been financial viability or practical feasibility. However, that does not mean that they should be deprived of their basic electricity needs.

These places could make good use of renewable energy especially solar energy, biomass, etc., which could lead to economic transformations thus making productive uses of renewable energy [9]. Productive use of renewable energy involves the utilization of energy – both electric, and non-electric energy in the forms of heat, and mechanical energy – for activities that enhance income and welfare. These activities are typically in the sectors of agriculture, rural enterprise, health, and education. Pumping of water for agriculture, agro-processing, lighting, information, and communications, as well as vaccine refrigeration, are examples of these activities [10]. Energy is very essential and is needed to excite production, income generation, and social development, and to reduce the critical health problems caused by the usage of fuelwood, dung, charcoal, and agricultural waste that is mostly used by individuals in off-grid communities. These communities are endowed with renewable energy resources that could be utilized for their benefits, but these resources are not looked into very critically. As demand for energy increases, the supply and demand gap keeps widening, and as such several alternative energy sources should be considered. Solar energy is one such promising source which has great potential to meet the growing demand. The location of West Africa, for instance, is under sunbelt [11]. A study conducted by Kearney et al. [12], showed that with adequate political support and deployment measurements in place, PV energy could be a sustainable and competitive technology that could provide up to 12% of the electricity demand of sunbelt countries by 2030 [11]. Immense efforts are presently being undertaken to promote renewable energy technologies, solar photovoltaics (PV) especially, as a means to achieving the Sustainable Development Goals (SDGs) agenda on the continent. Solar technology has become the most preferred form of renewable energy to promote productive use of energy [13]. Providing electricity to users especially in rural communities has always been guided by the cost involved in the process ignoring the entire benefits including meeting national objectives to the various stakeholders as extensively discussed by Lund et al. (2003) [14]. Several research papers have also discussed the feasibility of using solar PV to supply electricity to rural communities as against grid extension [15–19]. The argument has been surrounded by the cost and problems associated with grid extension such as distance from electricity grids, difficult terrain such as rivers or jungles, harsh weather conditions [8,10,11]. Renewable energy sources have a great potential to contribute to the sustainable development of specific places by providing them with a varied range of socioeconomic and environmental benefits. However, the existing literature has not comparably put much emphasis on socioeconomic impacts. These benefits include diversification of energy supply, improved regional and rural development opportunities, creation of the domestic industry, and employment opportunities [20].

Given this, it is important to evaluate the feasibility of energy projects using a wide array of specific and general national development objectives rather than narrow energy-sector objectives [21].

This research investigates the potential of a solar mini-grid system to meet the electricity needs of off-grid communities in Ghana, to promote productive use of energy. The analysis has been based on several factors such as economic impacts emanating from productive uses of electricity. It investigates the benefits of generating energy to rural areas and the country at large. The method used gives a better and deeper understanding of the case under study and thus the data used presents a reflection of the actual happenings in the community.

Materials and methods

The approach used in the research to estimate the potential energy demand, perform economic feasibility of solar resources towards energy generation, and assess the economic impact of productive use of energy towards the community was a survey and the use of RE software. The survey tool gives a better and deeper understanding of the case under study. The research employed both quantitative and qualitative methods in the analysis of data.

The data collection instruments that were used included; questionnaires, interview guides, focus group discussion, observation, and documentary review checklist. Computer-Aided Personal Interviewing (CAPI) techniques were used to collect data. CAPI uses handheld devices, in this case, a Samsung J3 and Samsung 6 edge to undertake the interviews.

Estimating energy demand from the data collected

Energy demand forecasting is an essential component in all power sector projects. The demand projections were based on the prospective electricity market, constituted by residential areas, commercial centers, hospitals, schools, and other general infrastructure, as well as for any future investments in those categories. It also relied on the respondent's decisions on the intended appliances they would want to use if they had electricity. Nevertheless, there are other issues such as

Table 1
Load estimates for low-class users.

Appliances	Power Rating per appliance (W)	Quantity	Duration per day (h)	Period of use per day	Power consumption (W)	Energy use per day (Wh)
Lights (CFL)	15	2	11	7 pm–6 am	30	330
phone charger	5	3	3	4 pm–7 pm	15	45
Radio	1	1	10	6 am–11 am, 5 pm –10 pm	10	100
TOTAL						475

Table 2
Load estimates for middle-class users.

Appliances	Power Rating per appliance (W)	Quantity	Duration per day (h)	Period of use per day	Total Power consumption (W)	Energy use per day (Wh)
Lights (CFL)	15	3	11	7 pm–6 am	45	495
phone charger	5	4	3	4 pm–7 pm	20	60
radio	10	1	10	5 am–10 am 5 pm –10 pm	10	100
fan	70	1	9	7 pm–4 am	70	630
TV	100	1	5	5 pm–10 pm	100	500
Decoder	25	1	5	5 pm–10 pm	25	125
TOTAL						1910

infrastructure and population growth, new business development, and new technologies that could stimulate a change in energy consumption.

The load or energy demand estimation was categorized into three sections namely:

- I **Residential users:** These covered structures in which the energy consumed were directly for purely household activities.
- II **Commercial users:** The commercial category covered structures in which the energy consumed was for activities that were not household activities including churches, schools, retail shops and all forms of businesses.
- III **Street lighting:** This category comprises structures that do not fall under commercial and residential categories but meant to illuminate the community.

Residential users load estimation

Based on an analysis of the responses from prospective end-users during the survey, the residential category was divided into various end-user classes for the segmentation of the end-users. This was proposed to guide in the load estimation process, it was evident during the survey that people just mentioned appliances they will not use when electricity is provided but for reasons unknown. Under each class of users, certain appliances were assumed to guide the process. The classes include; low-class users, middle-class users, and high-class users.

I Low-Class Users (low life-line end-users)

These end-users are assumed to consume up to 475 Watthours (0.475 kWh) per day or 13.5 kWh per month as shown in Table 1. At the present Uniform tariff rate determined by the PURC, these end-users will be paying about 4.5 Cedis per (\$0.9) month.

I Middle Class (Average income rural end-users)

These end-users are assumed to consume up to 1910 Watthours (1.91 kWh) per day or 57.3 kWh per month, this can be seen in Table 2. At the present Uniform tariff rate determined by the PURC, these end-users will be paying about 24 Cedis (\$4.82) per month.

I High-Class end Users

These end-users are assumed to use deep freezers and irons in addition to the basic appliances and consume up to 6735 Watthours (6.735 kWh) per day or 201.9 kWh per month as described in Table 3. At the present uniform tariff rate determined by the PURC, these end-users will pay about 108 Cedis (\$21.69) per month. These classes of end-users do not form the majority in most rural areas.

The final energy demand was calculated using the relation below;

Estimated energy for a class category= Number of prospective End-users *Estimated demand in the category

Based on the above relation, the energy demand calculations were performed as shown in Table 4

Table 3
Load estimates for High-class users.

Appliances	Power Rating per appliance (W)	Quantity	Duration per day (h)	Period of use per day	Total Power consumption (W)	Energy use per day (Wh)
Lights (CFL)	15	8	11	7 pm–6 am	120	1320
phone charger	5	4	3	6 pm–9 pm	20	60
fan	70	1	9	7 pm–4 am	70	630
TV	100	1	5	5 pm–10 pm	100	500
Deep freezer	400	1	8	1 pm–9 pm	400	3200
Decoder	25	1	5	5 pm–10 pm	25	125
Sound system	40	1	10	5 am–10 am, 5 pm–10 pm	40	400
Iron	1000	1	0.5	6 am–6:30 am	1000	500
					TOTAL	6735

Table 4
Baseline Energy Estimate.

Residential Tariff class	Number of prospective end-users	Estimated energy (kWh)
Low Class	25	11.875
Middle Class	38	72.58
High Class	4	26.94
TOTAL	67	109.875

Table 5
Load estimated under small-scale class.

Appliances	Power Rating per appliance (W)	Quantity	Duration per day (h)	Period of use per day	Power consumption (W)	Energy use per day (Wh)
Lights (CFL) indoor	15	1	3	6 pm–9 pm	15	45
lights (CFL) outdoor	20	2	12	7 pm–7 am	40	480
phone charger	5	1	3	2 pm–5 pm	5	15
radio	10	1	14	7 am–9 pm	10	140
					TOTAL	680

Commercial load estimation

Similarly, the commercial users were divided into three (3) categories namely; Small-scale users, Medium scale users and high scale users.

I Small-Scale Users

These commercial class users are typically for small groceries (provision) shops, small tailoring shops and other types of retail shops. These end-users use mainly lights at night, charge their phones, and use their radios during working hours. They consume up to 680 Wh (0.68 kWh) per day or 20.4 kWh per month. At the present Uniform tariff rate determined by the PURC, these end-users will pay about 29 Cedis (\$5.82) per month as detailed in Table 5.

I Medium User Class

These commercial tariff classes are typically for bigger groceries (provision) shops, tailoring and barbering shops, which do not use electric sewing machines or electric clippers, and other big retail shops. These end-users watch TV during working hours. They consume up to 1385 Wh (1.38 kWh) per day or 41.4 kWh per month shown in Table 6. At the present Uniform tariff rate determined by the PURC, these end-users will pay about 45 Cedis per (\$9.04) month.

I High Scale Users

These commercial class users are typically for cold stores, drinking bars and other shops selling cold drinks, which use deep freezers or fridge with freezers. These end-users also entertain themselves with TV during working hours. They consume up to 7200 Wh (7.20 kWh) per day or 216 kWh per month as detailed in Table 7. At the present Uniform tariff rate determined by the PURC, these end-users will pay about 186 Cedis (\$37.35) per month. Note that their fridges/freezers work for more hours than residential fridges because they usually opened more times during the day to serve customers. Those who use an ice-chest serving system usually save some energy.

A summary of the commercial load demand has been detailed using the above load estimates for various class of users as shown in Table 8.

Table 6

Load estimate for medium scale user.

Appliances	Power Rating per appliance (W)	Quantity	Duration per day (h)	Period of use per day	Total power consumption	Energy use per day (Wh)
Lights (CFL) indoor	15	2	3	6 pm–9 pm	30	90
Lights (CFL) outdoor	20	2	12	7 pm–7 am	40	480
phone charger	5	1	3	2 pm–5 pm	5	15
TV	1	1	8	1 am–9 pm	100	800
TOTAL						1385

Table 7

Estimated load for high user class.

Appliances	Power Rating per appliance (Watts)	Quantity	Duration per day (hours)	Period of use per day	Total power consumption (Watts)	Energy use per day (Watthours)
Lights (CFL) indoor	15	2	3	6 pm–9 pm	30	90
Lights (CFL) outdoor	20	2	12	7 pm–7 am	40	480
phone charger	5	2	3	2 pm–5 pm	10	30
TV	100	1	8	9 am–5 pm	100	800
Deep freezer or fridge with freezer	400	1	14	7 am–9 pm	400	5600
Decoder /Multi TV	25	1	8	9 am–5 pm	25	200
TOTAL						7200

Table 8

Commercial load demand per day.

Commercial/Institutions		Estimated energy demand (kWh)
Tariff class	Number of prospective end-users	
Small Scale	7	4.76
Medium Scale	9	12.222
High Scale	2	14.4
TOTAL	18	31.382

Table 9

Consumption of Streetlight.

Appliance	Duration per day (h)	Period of use per day	Power consumption	Energy use per day (Wh)
Streetlights	10	6:30–5:30 am	50	500.00

Street lighting

To ensure safety and clear illumination in the community and extend hours for activities to take place, streetlights should be installed in the communities. The number of streetlights to be factored in the estimation of the power and energy demand of the community was determined by assuming an average ratio of 1 street light to 3 houses in the community. The Energy consumption of each streetlight is shown in [Table 9](#)

The duration per day in hours is different from the period of use per day because the streetlights will not use 50 W throughout the night. At some point in time, the energy consumption will be managed using an automatic system.

As per the criteria stated above the proposed number of streetlights is equal to the number of households divided by 3
 Number of households = 85

Therefore the proposed number of streetlights is 29 and the estimated demand is 14.5 kWh per day

Daily load curve

To know the hour-to-hour variation of the load requirement of the Mini-Grid it is very important to study the load curve. The load curve shows the peak and baseload requirements. It helps to prepare the operational schedule, economic load dispatching between plants. The daily load curve is prepared by measuring the local loads of each plant and the sum up them to obtain the daily load curve of Mini-Grid (including daily average load curve, the peak / maximum load, minimum/baseload; and load factor). The Power demand is estimated for the Nkrankrom Community using a spreadsheet model on an hour (60 min period) developed with end-user patterns derived from the survey in the communities. This is purposely for modeling using the HOMER software. The load curves as shown in [Tables 10–13](#).

Table 10

Cumulative power demand for each hour profile during the day for 12 am – 6 am.

Tariff class/ Time of the day	12 am–1 am	1 am–2 am	2 am–3 am	3 am–4 am	4 am–5 am	5 am–6 am
Residential	6290	6290	6290	6290	5730	5870
Commercial	1520	1520	1520	1520	1520	1520
Streetlights	2610	0	0	0	2610	1305
Total in Watts	10,420	7810	7810	7810	9860	8695
Total in KW	10.42	7.81	7.81	7.81	9.86	8.695

Table 11

Cumulative power demand for each hour profile during the day for 6 am–12 noon.

Tariff class/ Time of the day	6 am–7 am	7 am– 8 am	8 am–9 am	9 am–10 am	10 am–11 am	11 am–12 am
Residential	1400	1400	1400	1400	1260	0
Commercial	1520	900	900	5550	5190	1830
Streetlights	0	0	0	0	0	0
Total in Watts	2920	2300	2300	6950	6450	1830
Total in KW	2.92	2.3	2.3	6.95	6.45	1.83

Table 12

Cumulative power demand for each hour profile during the day for 12noon – 6 pm.

Tariff class/ Time of the day	12 noon–1 pm	1 pm–2 pm	2 pm–3 pm	3 pm–4 pm	4 pm–5 pm	5 pm–6 pm
Residential	0	400	400	400	2920	15,520
Commercial	1830	7430	3640	3640	7640	2500
Streetlights	0	0	0	0	0	0
Total in Watts	1830	7830	4040	4040	10,560	18,020
Total in KW	1.83	7.83	4.04	4.04	10.56	18.02

Table 13

Cumulative power demand for each half-hour profile during the day for 6 pm– 12 am.

Tariff class/ Time of the day	6 pm–7 pm	7 pm–8 pm	8 pm–9 pm	9 pm–10 pm	10 pm–11 pm	11 pm–12 pm
Residential	15,560	19,330	19,330	8690	6290	6290
Commercial	7310	8830	8830	1520	1520	1520
Streetlights	0	2610	2610	2610	2610	2610
Total in Watts	22,870	30,770	30,770	12,820	10,420	10,420
Total in KW	22.87	30.77	30.77	12.82	10.42	10.42

Modeling and simulation in HOMER

The idea behind the use of the HOMER software is to perform the economic feasibility of solar resources towards energy generation. The software is utilized to investigate the possibility of using the site resources to meet the energy demand of the community under study thus Nkrankrom. The program considers scenarios and simulates them to meet demand. Situations can be analyzed as either grid-tied or standalone, and their emissions levels estimated. HOMER allows the user to make inputs of RE generation to the required demand. This allows the user to analyze potential small-scale power systems before implementation. It also allows comparing many different design options with regards to their technical and economic benefits [22]. The simulation procedure serves two purposes. Foremost, it defines whether the system is feasible. HOMER selects the system to be feasible if it can sufficiently serve the electric and thermal loads and satisfy any other constraints imposed by the user. Next, it estimates the life-cycle cost of the system. The program models a particular system configuration by undergoing an hourly time series simulation of its operation over one year. HOMER does this throughout the year by considering one hour at a time and matching the available renewable power to the electric load utilizing both resource data and load data. A decision is then made on what to do with extra renewable power in times of excess, or how best to generate or purchase from the grid in times of shortfall as shown in Fig. 1 and Fig. 2. In Fig. 2, for instance, the battery bank absorbs energy when the PV power output exceeds the load and discharges energy when the load exceeds the PV power output. This, however, does not take into account changes such as load growth or the deterioration of battery performance with aging which occurs over time [23]. Fig. 3 below shows the system modeled in HOMER.

Load input

The program requires users to input load data for simulation. The required load data must be 24 h of hourly energy consumption of the electrical load in kilowatts intended to meet. A load in HOMER is known as electric or thermal energy. Meeting the demand of the load loads is the purpose of the existence of micropower systems, therefore modeling of the load marks the start of the process. HOMER synthesizes hourly data from average daily load profiles for the entire year [24].

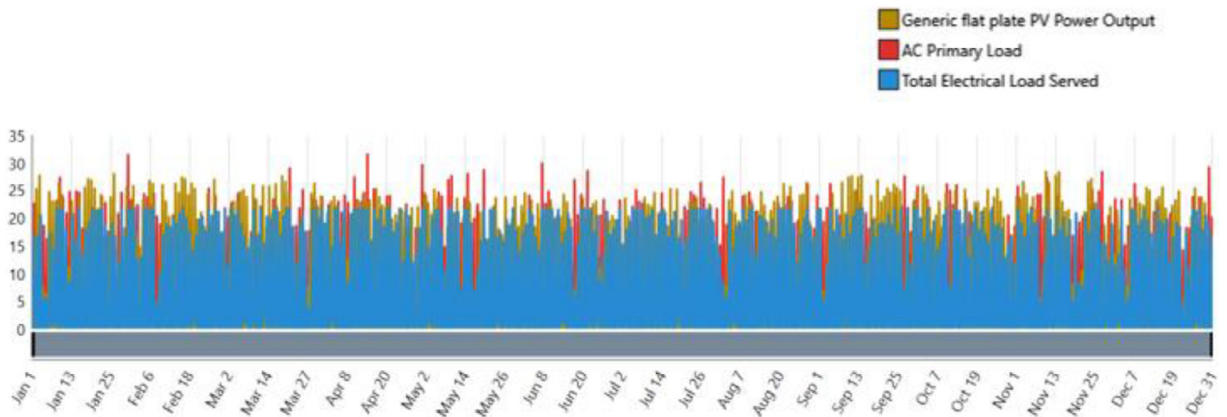


Fig. 1. How Homer evaluates PV productions and electricity demands.

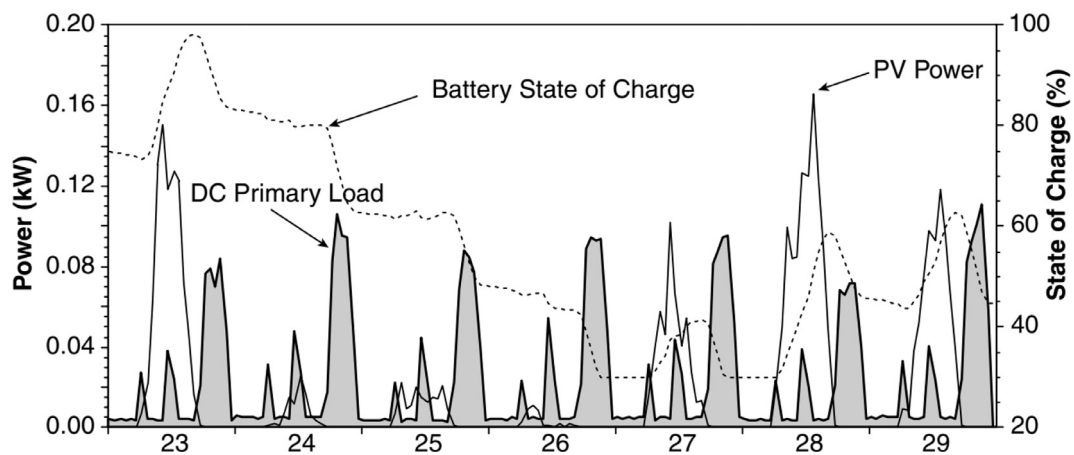


Fig. 2. Sample hourly simulation results [23].

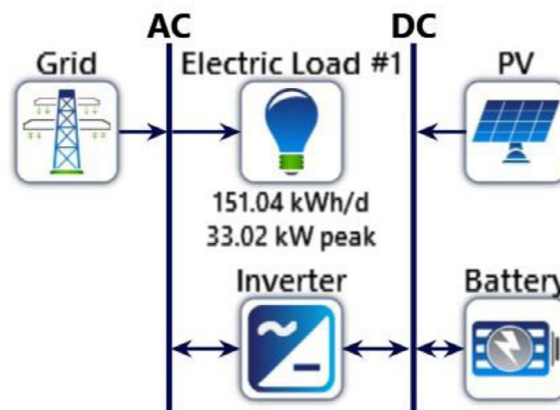


Fig. 3. HOMER Model of the Hybrid Power System.

Resources input

This section requires the site's solar resource data to model, simulate, and analyze a system's performance. The solar resource data of the site was automatically downloaded using the HOMER software. The site receives annual daily averaged global solar irradiation ranging from 5.30 kWh/m² to 5.84 kWh/m². The solar radiation data used in this study is satellite data obtained from NASA Surface Meteorology [15]. This choice has been made because there was no publicly available ground measured solar radiation data for the selected site. HOMER uses the location for several aspects of the PV power

Table 14

Productive use of energy activities.

Productive Use of Energy (PUE)	Estimated number
Production of ice water, ice blocks and ice-cream for sale using Deep freezers	2
Operation of a commercial video center using Television and Decoder	2
Hairdressing, barbering and tailoring shops using hair dryers, electrical clippers and electric powered sewing machines.	1
Drinking spots/ drinking bars using fridges, TV, Sound system	2
use of water pumps for crop irrigation using Electric water pump	1
Maize processing using Maize processors	1
corn mills using Electric mills	2
Bakeries using Electric ovens	2

calculation. It builds a set of 8760 h of solar radiation values, for the year using the Graham algorithm, which results in a data structure that has a representative of day-to-day and hour-to-hour variability and auto-correlation [23].

Load input

The program requires users to input load data for simulation. The required load data must be 24 h of hourly energy consumption of the electrical load intended to meet. [24].

Batteries

The input parameters of batteries are the cost and the number of batteries required. The following are also defined for the batteries: nominal capacity, nominal voltage, round trip efficiency, minimum state of charge, float life, maximum charge rate, maximum current, and lifetime throughput. In this study, Trojan SAGM 12 135 was used because these batteries are an inexpensive option and have a low maintenance requirement. In a practical installation, a smaller number of larger batteries may be preferable, and economic analysis performed by HOMER would not be significantly affected by this difference [22]. The capital and replacement cost of one battery is taken as US\$339 and US\$300.00 respectively and the operational and maintenance cost is taken as US\$5/year [25]

Converter

According to the survey conducted, the proposed appliances by the respondents were AC electrical loads and so there is the need to include a converter to the model. The unit capital and the replacement costs of a 5 kW converter was taken as US\$867 and US\$800.00. The inverter efficiency was given as 96%. The lifetime of a unit is taken as 15 years since the selected manufacturer gives a 10-year warranty [17]. No particular size was selected because HOMER was set to select the best sizes.

Estimates for grid extension

The grid extension cost in Sub-Saharan Africa has been estimated averagely by Longe et al. [26] as \$20,000/km for 11 kV line cost. The cost used for this design is \$10,607.3/km, which is within the range of cost. The cost is less because the site/area under study is of good terrain. The cost of extending the grid to mountainous areas is generally higher than in places of good terrain.

Other inputs

Other inputs needed for HOMER to complete its computation include; economics, constraints, emission, system control input parameters. The economic parameters that need to be set to determine NPC are annual real interest rate, project lifetime, system fixed capital cost, system fixed operating and maintenance (O & M) cost, and capacity shortage penalty. The lifetime of the proposed system is assumed to be 25 years, which is typical; the annual real interest rate is taken to be 15.2% from [27]

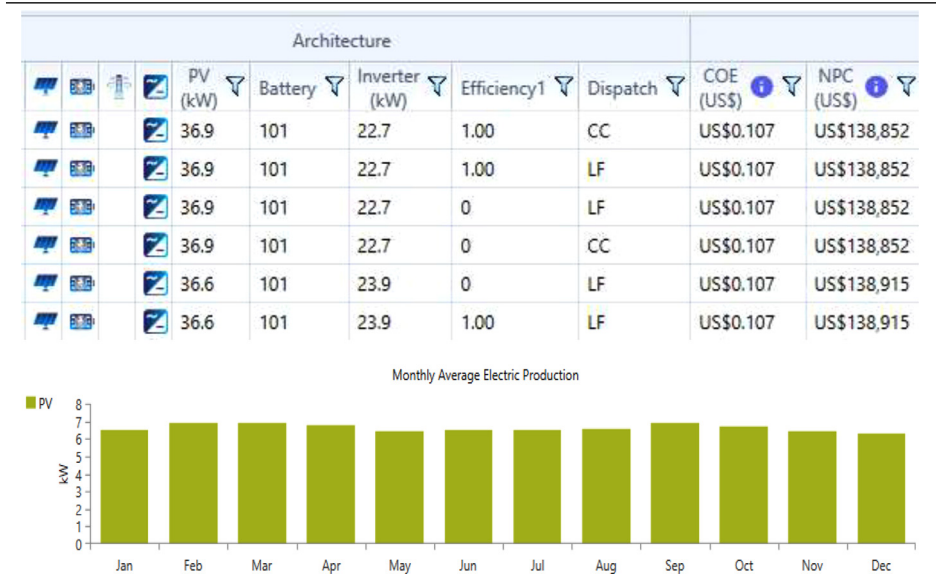
Economic impact of productive use of energy towards nkrankrom community

As expected in un-electrified rural communities, the main application of electricity (from dry cell batteries, solar PV systems, Car batteries, etc.) in the communities visited is for “unproductive” uses. Productive Uses of Energy (PUEs) ensures that the investment in the electrification system makes more economic sense. Based on the survey data, expected productive applications of the electricity to be supplied over 10 years is estimated as follows in Table 14;

Results and discussion

This section explains the results obtained from the study to achieve the stated objectives. It includes graphs, charts, and tables from the software (HOMER).

Table 15
Optimization results from HOMER.



HOMER simulations

HOMER presents a series of models that could meet the load demand under the overall settings. It does this by putting the best model on top of the list of possible solutions. The results are shown in Table 15. It shows the results for the overall system configurations reported by HOMER after the optimization simulation. The most cost-effective configuration for the proposed mini-grid is the PV/Battery/Converter combination having the least total NPC of \$138,852 with \$0.107/kWh cost of electricity (COE). This optimal system is composed of 36.9 kW solar PV, 101 pieces of 135Ah, 12 V batteries and a 22.7 kW converter. The average electricity produced monthly from the proposed mini-grid is shown in Fig. 4.

The solar resource of the people of Nkrankrom is capable of meeting the electrical requirements of the people. The minimum distance of unelectrified areas in Nkrankrom from the grid is about 36 km from the grid, but the break-even distance of this design was found as -1.11 km, which makes it more economical to utilize the standalone solar photovoltaic system as against the grid extension. The break-even distance suggests that it would only be feasible or make economic sense than the standalone system if the distance from the next available grid is -1.11 km.

Mini-grid design cost summary

The HOMER software provides the summary cost of the proposed system to meet the load demand of the community. The total NPC of each component in the optimal system design is shown in Fig. 5. The costs considered include capital costs, installation cost, equipment cost, and replacement cost. Cash flow calculations did not include discounts, government funding or any form of local or international subsidies, which could change the dynamics of the payback period of the project. In the cost summary, as shown in Fig. 5, it is seen that the total cost for PV, battery, and converter for the optimal system design is \$22,925, \$105,810, and \$10,117 respectively.

Breakeven distance for grid extension

The breakeven distance or Electric Distance Limit (EDL) between standalone mini-grid and grid extension in this analysis was found to be -1.11 km. This implies that the Nkrankrom solar mini-grid would be more economical and optimal for the unelectrified areas with similar properties. Most unelectrified areas in Sunyani West District are more than 36 km from the grid. At such distances, the cost of grid extension would likely far exceed the cost of the proposed standalone mini-grid. The distance from Nkrankrom to where a substation is being constructed is 36 km. The absence of transmission lines in rural areas makes the grid extension economically unviable. The cost of grid connection keeps increasing however, the cost of renewable energy products in the market keeps increasing which makes a better case for rural electrification through standalone mini-grid.

Summary of economic analysis

The economic analysis of the solar mini-grid simulated using HOMER software shows that the site resource would be able to economically generate energy to meet the demand which will cost \$0.1068/kWh to recover the total cost of the system

Architecture										
			PV (kW)	Battery	Inverter (kW)	Efficiency1	Dispatch		COE (US\$)	NPC (US\$)
			36.9	101	22.7	1.00	CC		US\$0.107	US\$138,852
			36.9	101	22.7	1.00	LF		US\$0.107	US\$138,852
			36.9	101	22.7	0	LF		US\$0.107	US\$138,852
			36.9	101	22.7	0	CC		US\$0.107	US\$138,852
			36.6	101	23.9	0	LF		US\$0.107	US\$138,915
			36.6	101	23.9	1.00	LF		US\$0.107	US\$138,915

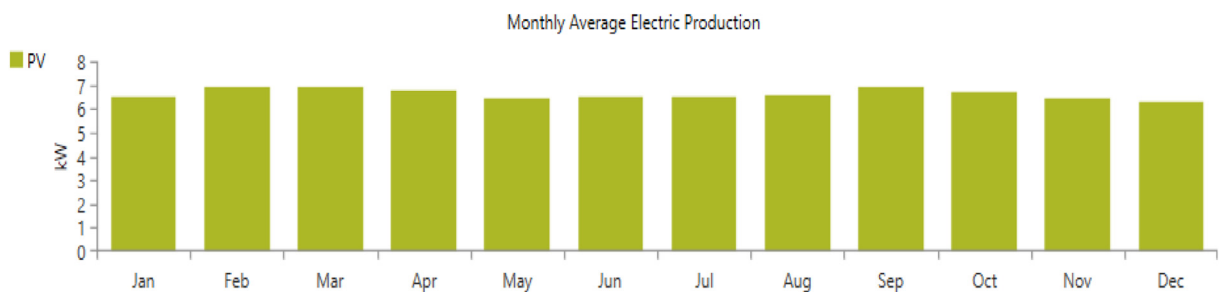


Fig. 4. Monthly average electric production from PV.



Cost Summary

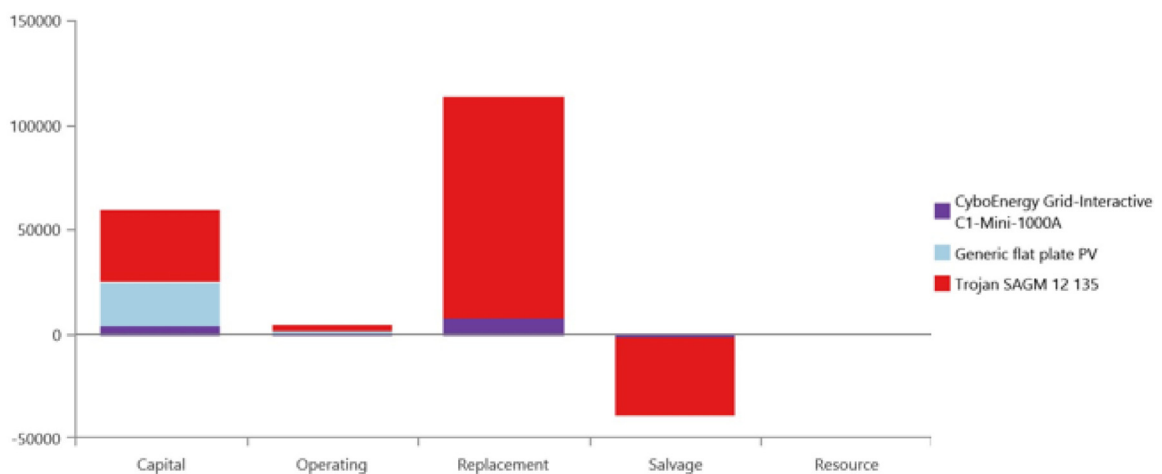


Fig. 5. Cost summary of the proposed system.

at the end of the 25 years lifespan of the project/system. The cost of electricity in Ghana per kWh is \$0.124 (\$1=GH¢4.98). This means that the system cost less as compared to grid extension. Also, the breakeven distance or Electric Distance Limit (EDL) between standalone mini-grid and grid extension in this analysis was found to be –1.11 km meanwhile the distance between the community under study and the next available town where the grid was extended is 36 km. This makes the grid extension more expensive compared to the proposed standalone system.

Expected impact of the solar mini-grid on economic and social life of the people of Nkrankrom

Most developing countries to accelerate their rise and integration in the world economy turn to increase investment in rural electrification. They do this by building on a strong agricultural base and on vast natural resources, industrialization and the service sector to generate jobs and welfare for their people. It is important to note that it would be impossible without access to affordable and reliable energy. Energy access has unavoidably become pre-requisite for local economic growth such as small companies, local industrialization and agricultural upscaling [28].

Energy supply improvement in rural communities will safeguard enhancement in the livelihood of the rural population and the achievement of sustainable economic growth thus through the promotion of productive use of energy. Ghana's rural economy will be transformed when agriculture, agro-business, small and medium industries and enterprises, and commercial establishments in the rural sector have access to and use modern energy. This will increase significantly the value added to rural production and will stimulate investment, job-creation, narrow the gender gap, and revenue generation throughout the rural segment [29].

Energy access is an important element to empower people. The non-availability of electricity supply in schools or offices (for light, use of computers), clinics (refrigeration of medicine, disinfection of tools), for water supply facilities, for engines in workshops, for telephone chargers, televisions, radios, and household appliances can render daily life challenging [30]

Conclusions and limitations of the study

Conclusions

This study has shown that for an unelectrified locality like Nkrankrom in the Bono region of Ghana, a solar mini-grid is cost-effective than grid extension. The optimal system configuration includes PV/Battery/Converter. The Levelized Cost of Energy for such a system would be \$0.107/kWh compared to \$0.124/kWh (\$1 = GH¢4.98) if the area is to be connected to the grid. Due to the use of renewables, the system would result in a safer environment. The results are robust to uncertainties in resource availability, component cost, and some load growth. The study suggests that mini-grids merit serious consideration, at least on techno-economic grounds, as an electrification option for off-the-grid unelectrified villages in Ghana and other developing countries of the world. According to the breakeven distance or Electric Distance Limit (EDL) between standalone mini-grid and grid extension, it would be cost-effective to invest in mini-technology.

Renewable energy sources (RES) mini-grid also has certain environmental benefits as it will contribute to a reduction in pollution from wood-burning and the preservation of the forests. Also, the RES mini-grid would reduce some health and safety risks associated with the use of wood, paraffin, and candles for lighting, heating, and cooking.

In many of the literature reviewed there has been the creation of new business ideas that were not in the village before the access of electricity, business-like television showroom, game centers and CDs lending shops. These are not different from the responses received from the community members on expected transformations that are likely to take place in the community.

It has been observed that the presence of electricity would be a huge help to all age groups such as students, youths, and people of old age including women and men living in the community. Students would have longer hours of studying during the night hence their performance would increase. The results show that Solar has the potential to meet the electrical needs of the community.

In conclusion, solar mini-grid has vast benefits to the rural communities and the country at large.

Limitation of the study

Although the results obtained satisfies the objectives of the project, grid power in principle, would be better able to handle a very large load increases. The long-term feasibility of off-grid depends on other factors such as availability of repair facilities, availability of maintenance personnel, communal influences, insurance, security, land cost, etc., which were not considered in this study. Besides, the study did not look at forecasting load demand, which is very important in mini-grid management.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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